

FIRST 3D OBJECT / FULL GUIDE

SKYBOUND

Your first 3D expedition



A real model. An intentional light. A photographed world.

32 core steps + 6 optional extensions

Blender 4.5 LTS → transparent PNG → Photoshop

Import • texture • camera • lighting • render • composite

What you are making

The core challenge

Create a scene with one textured 3D balloon, light it deliberately, render it with transparency, and integrate it into the supplied landscape. Begin with the core, then choose an extension that strengthens your story.

SKYBOUND / Core first. Storytelling second.

Source-model teaching composites • no AI-generated replacement model



CORE / one 3D balloon



EXTENSION / a campaign world

The optional effects do not replace evidence of importing, texturing and rendering.

Core on the left; optional campaign on the right. These are source-model teaching composites.

A genuine workflow update

Photoshop's legacy 3D features were removed in July 2024 (A1). Substance 3D Viewer and its Photoshop workflow were subsequently discontinued (A2). The core uses Blender for genuine 3D and Photoshop for the rendered PNG. It does not require obsolete Photoshop 3D tools or generative credits.

What the references can and cannot prove

The model was parsed, adapted and rendered in VTK 9.6.2. The diagrams and source-based composites are teaching references, not Blender or Photoshop screenshots. Blender and Photoshop were not available for an end-to-end software run here. A teacher must complete the installed-version preflight before independent student use.

Teacher preflight is required before independent student use. All numeric positions, powers, brush sizes and corrections are starting values for the prepared assets. Render engines and installed software can differ. Match the visual checkpoint rather than forcing a harmful setting.

Choose a clear path

A • Meet the model | Steps 01-08 | pp. 7-14

Set up files, navigate, identify parts, parent and transform.

B • Surface and camera | Steps 09-14 | pp. 15-20

Connect the artwork, inspect UVs, control roughness and frame the camera.

C • Light and render | Steps 15-21 | pp. 21-27

Light the model, export PNG alpha, compare real angles and save the scene.

D • Build the composite | Steps 22-32 | pp. 28-38

Place the render, match tones, add mist, revise and submit.

E • Extend the story | Steps E1-E6 | pp. 39-44

Custom branding, fleet, mirage city, controlled storm and campaign type.

After the walkthrough

Visual reference cards: pp. 45–47. Troubleshooting: pp. 48–49. Reflection: p. 50. Suggested assessment: p. 51. Teacher preflight: p. 52. Credits and sources: pp. 53–55.

Suggested time

Five to six 70-minute classes for first-time learners, plus optional extension time. This is a planning estimate; local rendering speed and student needs vary.

Checkpoint routine

Read SELECT FIRST. Do the numbered actions. Compare CHECK. Use FIX when the result differs. Record the decision, then mark the step complete. Checkboxes are progress aids, not grades.

Know your assets

Balloon_Classroom.obj + .mtl [\[Core\]](#)

Prepared mesh and four starting materials. Keep these files together. About 14 MB uncompressed OBJ; roughly 186,000 faces.

Skybound_Texture.png [\[Core\]](#)

1571 × 887 new texture; aligned to the retained UVs. Image connection is taught explicitly.

Landscape.jpg [\[Core\]](#)

2000 × 1333 supplied photograph, renamed for clarity. The working canvas matches it.

Texture_Blank.png [\[Extension\]](#)

Matching empty layout for an original fictional expedition identity.

Balloon_A_Front.png / Balloon_B_Turned.png [\[Support only\]](#)

1200 × 1200 transparent VTK renders. Not a substitute for the learner's own Blender evidence.

Mist_Support.png [\[Support only\]](#)

2000 × 1333 prepared soft-painted overlay. The core also teaches painting it yourself.

CityOutline.png / Clouds_Storm.jpg / Clouds_Warm.jpg [\[Extension\]](#)

Supplied source elements. Storm and city instructions are optional; warm clouds remain an alternative source.

Original reference / original decal [\[Reference\]](#)

Supplied historical composite and its original artwork, retained for comparison.

original_model/balloon.obj + .mtl [\[Teacher archive\]](#)

Full untouched raw model in the complete classroom ZIP only. It is heavier and has many groups; do not use it for the first import.

The self-contained HTML includes a starter ZIP download. The complete classroom ZIP also retains the untouched raw model and editable lesson text. No .blend is supplied; you build and save your own.

Controls you will actually use

Select first

Choose the named object in Blender's Outliner, or the named layer/mask in Photoshop. Read the selection line before doing the actions.

Blender: inspect

Middle-drag or navigation gizmo = orbit. Shift+middle-drag = pan. Wheel = zoom. View > Frame Selected finds the selected object.

Blender: transform

G = move, R = rotate, S = scale. Then X, Y or Z constrains the axis. Type a number and Enter. Esc cancels the operation.

Blender: modes

Tab switches Object/Edit Mode. N opens the viewport Sidebar. F3 searches commands. Keep the pointer over the correct editor.

Blender: camera/render

View > Cameras > Active Camera (Numpad 0). Render > Render Image (F12). Then Image > Save As saves the actual PNG.

Photoshop: main keys

V = Move, B = Brush. Ctrl+T on Windows / Cmd+T on Mac = Free Transform. Ctrl/Cmd+S saves the document.

Photoshop: masks

Click the mask thumbnail first. White reveals, black hides. Ctrl/Cmd+I inverts a selected mask. A clipping mask restricts an adjustment to the image beneath.

Keyboard differences

The guide assumes Blender's default keymap. Blender's G/R/S keys work on either platform; modifier shortcuts may differ with custom keymaps or OS shortcuts. Use the written menus when a shortcut is intercepted. Function keys may require Fn on a laptop.

The starting-settings card

Prepared model Balloon_Classroom.obj + matching .mtl; 4 named objects; Z-up; 12 scene units high.

OBJ import Forward Y / Up Z / Scale 1 / Clamp 0. Split by Object ON, Group OFF. Validate Meshes ON.

Controller BALLOON_CTRL: Location 0,0,0; Rotation 0,0,0; Scale 1,1,1. Parent the four meshes with Keep Transform.

Envelope material Image Texture Color → Principled Base Color. Active UV map. sRGB image. Roughness 0.65 / Metallic 0 / Alpha 1.

Camera_Render Location (0, -26, 6); Rotation (90°, 0°, 0°); Orthographic; Scale 14.4.

Key_Area Location (-6, -8, 13); Rotation (59.04°, 0°, -36.87°); 1800 W; Square Size 5; pale warm colour.

Fill_Area Location (5, -4, 7); Rotation (90°, 0°, 51.34°); 500 W; Square Size 6; neutral white.

Render/export EEVEE; 1200 × 1200; 100% final / 50% test. Film Transparent ON. Save PNG/RGBA/8-bit; Save As Render ON.

View transform Render Properties > Color Management > AgX. Keep view settings consistent between tests.

Photoshop canvas 2000 × 1333 pixels; RGB/8; sRGB. Core render centre approximately (1430,650), using its full 1200-pixel canvas.

Core corrections Background saturation -16. Balloon Curves 128→120. Local highlight 128→144 on a masked/clipped adjustment. Warm Photo Filter 85, Density 8%. These are starting points, not grading criteria.

Positions are Blender X/Y/Z scene units; rotations are degrees. Photoshop positions are canvas pixels. These recipe values apply to the prepared classroom model only.

Set up your expedition

SELECT FIRST Your downloaded starter ZIP — not the original finished PSD.

02 / Know the files before you start

Teaching diagram • not a software screenshot

Balloon_Classroom.obj

Shape + UV coordinates

Balloon_Classroom.mtl

Four starting material colours

Skybound_Texture.png

Artwork for the curved envelope

Landscape.jpg

The 2D scene behind the balloon

Balloon_A_Front.png

Optional prepared render support

Student_Skybound.blend

Your editable 3D scene — you create it

The classroom file names are intentionally different from the raw original.

1. Extract the starter ZIP into a new folder named Lastname_Skybound. Do not work inside the compressed ZIP. Keep the OBJ and MTL together in its starter folder.
2. Create three empty folders inside your project: 3D_Work, My_Renders and Photoshop_Work. Keep the supplied texture and landscape files in starter.
3. Open Core_Final.jpg in the guide. Identify the single large balloon, the light arriving from the upper left, and the small amount of mist crossing its lower section.
4. Write a one-sentence goal: “I will make a 3D balloon belong in a photographed landscape by ...”. Choose the core route first; the city, fleet and poster are extensions.
5. Confirm that your teacher has preflighted Blender 4.5 LTS and Photoshop desktop on this computer. A three-button mouse helps, but the View menus provide alternatives.

WHY Separating sources, editable work and exports prevents accidental overwrites. This project teaches genuine 3D decisions, not just placing a supplied PNG.

CHECK You can locate Balloon_Classroom.obj, Balloon_Classroom.mtl, Skybound_Texture.png and Landscape.jpg without opening a ZIP.

FIX Missing companion files? Extract the entire starter ZIP again. The large original balloon.obj is an archive reference, not the classroom import.

Evidence: Record your goal and any prepared support you expect to use.

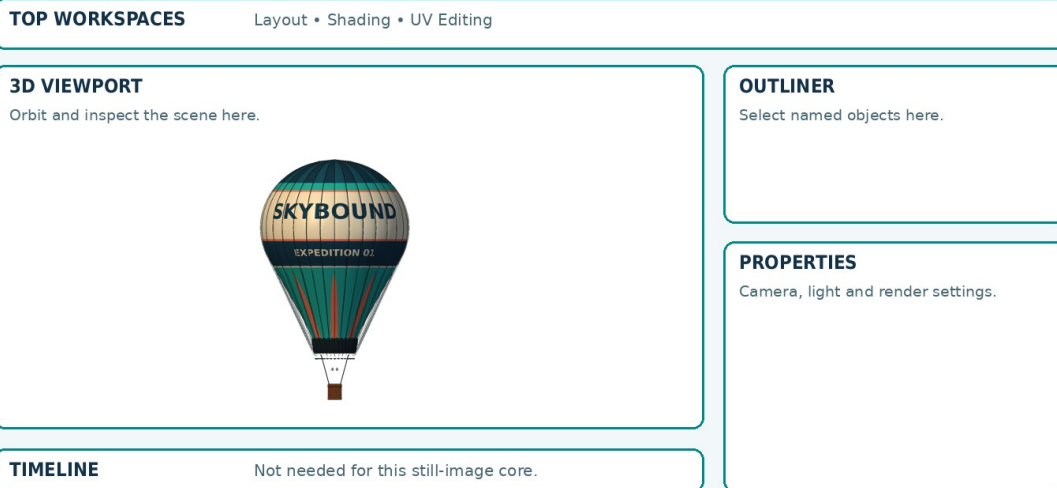
Tool references: B1, B2 · Full links on pp. 54–55.

Find your way around Blender

SELECT FIRST A new General scene; save any existing personal work before replacing it.

START HERE / The main Blender editors

Orientation diagram only • not a screenshot • workspace sizes may differ



Hover over the correct editor before using a shortcut. Look at the selected object first.

Editor locations are a labelled orientation diagram, not a Blender screenshot.

1. Choose File > New > General. Click outside the splash screen. Select the Layout workspace at the top.
2. Locate the large 3D Viewport, the Outliner at the upper right, and the Properties editor below it. These are different editors: shortcuts act in the editor under your pointer.
3. Move the pointer into the 3D Viewport. Confirm the mode menu says Object Mode. Press A to select the startup cube, camera and light; press X, then confirm Delete.
4. Choose File > Save As. Save Lastname_Skybound.blend in 3D_Work. You will save again after importing, texturing and rendering.
5. Leave the central 3D cursor at the world origin. To reset it, use Shift+S > Cursor to World Origin with the pointer over the viewport.

WHY A clean scene makes each object you add understandable. The Outliner is the scene inventory; Properties changes the selected object or the scene.

CHECK The Outliner contains an empty collection, and there is no startup cube, camera or light. You have saved a .blend file.

FIX X did nothing or deleted a node? Move the pointer over the 3D Viewport. An unsaved unrelated project should be saved separately before File > New.

Tool references: B3 • Full links on pp. 54–55.

Import the four-part classroom model

SELECT FIRST File > Import > Wavefront (.obj).

03 / Import the classroom model — not the raw original

Teaching diagram • not a software screenshot



Import settings

File > Import > Wavefront (.obj) Choose
Balloon_Classroom.obj. Scale 1.000. Forward Y.
Up Z. Split by Object ON. Split by Group OFF.

Target: upright / 12 units tall / 4 parts

Import settings apply only to Balloon_Classroom.obj, not the untouched Y-up original.

1. Navigate to starter and choose Balloon_Classroom.obj. Show the import options on the right of the file browser.
2. Set Scale to 1.000 and Clamp Bounding Box to 0 (disabled). Set Forward Axis to Y and Up Axis to Z. These settings preserve this prepared Z-up model's coordinates.
3. Enable Split by Object. Disable Split by Group and Vertex Groups. Keep Validate Meshes enabled. Click Import Wavefront OBJ and allow the import to finish.
4. With the imported parts selected, choose View > Frame Selected in the 3D Viewport. The envelope should be upright above the basket.
5. Check the Outliner for Envelope, Rigging, Basket and Hardware. Do not rename them yet; the next steps use these names.

WHY OBJ supplies the mesh and UV coordinates, while the matching MTL supplies starting materials. It does not bring in a complete camera-and-light scene. Our prepared file keeps the original envelope and UV layout while simplifying and organising supporting parts.

CHECK Four mesh objects appear. The complete model is about 12 scene units high, with its basket base near $Z = 0$.

FIX Sideways model? Undo the import, then repeat with Up Z. Facing backwards later? Confirm Forward Y, not -Y. Extra parts? Use the classroom OBJ, Split by Object ON and Group OFF.

Tool references: B2 • Full links on pp. 54–55.

Orbit, pan and zoom without moving the model

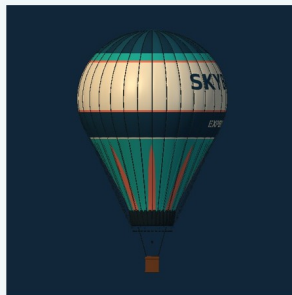
SELECT FIRST All four imported parts; pointer in the 3D Viewport.

04 / Orbit changes your view — not the model

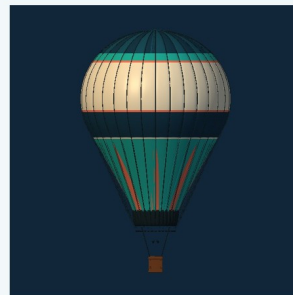
Teaching diagram • not a software screenshot



FRONT



SIDE



BACK

Middle mouse: orbit • Shift + middle mouse: pan • Wheel: zoom

No numpad? Use View > Frame Selected and View > Viewpoint > Front.

Three genuine source-model render angles. The HTML also includes a 12-view turntable.

1. Drag with the middle mouse button to orbit around the balloon. Hold Shift while middle-dragging to pan. Scroll the wheel to zoom.
2. Choose View > Viewpoint > Front, then Right, then Top. With a numpad, the corresponding shortcuts are Numpad 1, 3 and 7.
3. Return to Front. Use View > Frame Selected when you lose the model; press Home to frame everything in the scene if nothing useful is selected.
4. On a mouse without a middle button, use the viewport navigation gizmo: drag the axis control to orbit and use the hand and magnifier controls to pan and zoom.
5. Observe which parts overlap in each view. Write the difference between orbiting the viewport and rotating the balloon itself.

WHY Orbiting changes only your inspection view. Object rotation changes the scene and therefore the final camera render. Confusing those actions is a common first-3D problem.

CHECK You can inspect the front and side and return to Front without changing any object transform values.

FIX The balloon moved instead? Undo once; you probably dragged with a transform tool. Use middle mouse or the navigation gizmo, not a move arrow.

Evidence: Demonstrate orbit, pan and Frame Selected to your teacher or partner.

Tool references: B3 • Full links on pp. 54–55.

Identify the pieces you will control

SELECT FIRST Envelope in the Outliner.

05 / Four parts, one controller

Teaching diagram • not a software screenshot



Envelope

The fabric surface. Select this for the image texture.

Rigging

Lines, suspension supports and the dark skirt.

Basket

The carrier below the envelope.

Hardware

Small burner and internal detail.

BALLOON_CTRL parents all four.

Four groups reorganised from the supplied model; the full preparation record is in teacher/.

1. Click Envelope in the Outliner. Notice that the fabric becomes selected in the viewport. Select Basket, then Rigging, then Hardware to compare their roles.
2. Use the eye icon beside Envelope to hide it temporarily. Inspect the basket, lines and small hardware. Click the eye again to restore the fabric.
3. Click Envelope and press N over the viewport to open the Sidebar. In Item, inspect its Location, Rotation and Scale. Do not edit these values yet.
4. Choose View > Viewpoint > Front. Confirm that all four parts are visible again. Save the .blend file.

WHY Selection determines what a command affects. The image texture belongs on Envelope; painting a material on Basket will not decorate the fabric. Small internal details do not need their own beginner lesson.

CHECK You can point to the four named pieces and identify the fabric independently. No part is left hidden.

FIX Missing ropes or basket? Restore the eye icons in the Outliner. If a collection is hidden, restore that collection too. Do not duplicate a “missing” object before checking visibility.

Tool references: B2 • Full links on pp. 54–55.

Parent the balloon to one controller

SELECT FIRST The model only — there must not yet be a camera or light in the scene.

06 / Parent the parts — keep the camera independent

Teaching diagram • not a software screenshot

BALLOON_CTRL [Empty]

Move, rotate and scale this object.

Envelope

Rigging

Basket

Hardware

Camera_Render

Separate from BALLOON_CTRL.

Key_Area + Fill_Area

Separate from BALLOON_CTRL.

Why it matters

Rotating only the fabric leaves the basket behind. Parenting keeps the entire balloon together.

The hierarchy is a diagram of the intended scene structure.

1. Over the viewport, use Shift+S > Cursor to World Origin. Choose Add > Empty > Plain Axes. Rename the new Empty BALLOON_CTRL using F2.
2. The newly added Empty is the active object. With the pointer over the viewport, press A to select all five objects while keeping the Empty active.
3. Check the active object name is BALLOON_CTRL. If it is not, deselect everything, select the four meshes, then Shift-select the Empty last.
4. Use Object > Parent > Object (Keep Transform), or Ctrl+P and choose Object (Keep Transform). Expand BALLOON_CTRL in the Outliner.
5. Confirm that Envelope, Rigging, Basket and Hardware are children. Leave all controller transforms at their defaults: location 0, rotation 0, scale 1.

WHY Parenting links the parts without merging their meshes or materials. You can rotate the whole balloon while still editing only its fabric. Cameras and lights must stay independent.

CHECK The four meshes are indented below BALLOON_CTRL in the Outliner, and none has jumped position.

FIX Basket stayed outside the hierarchy? Undo and repeat with all four meshes selected before the Empty. Fabric jumped? Use Keep Transform, not a different parenting operation.

Tool references: B4 • Full links on pp. 54–55.

Practise transformations, then reset

SELECT FIRST BALLOON_CTRL in Object Mode; not an individual child mesh.

06 / Parent the parts — keep the camera independent

Teaching diagram • not a software screenshot

BALLOON_CTRL [Empty]

Move, rotate and scale this object.

Envelope

Rigging

Basket

Hardware

Camera_Render

Separate from BALLOON_CTRL.

Key_Area + Fill_Area

Separate from BALLOON_CTRL.

Why it matters

Rotating only the fabric leaves the basket behind. Parenting keeps the entire balloon together.

G moves, R rotates and S scales. X, Y or Z constrains the active transformation.

1. With the pointer in the viewport, press G, then X, type 2 and press Enter. The entire balloon moves two units along X. Undo with Ctrl+Z.
2. Press R, then Z, type 30 and press Enter. Inspect the changed face of the balloon. Undo to return to the starting rotation.
3. Press S, type 0.8 and press Enter. All parts should shrink together. Undo again.
4. Open the Sidebar with N. Verify BALLOON_CTRL has Location (0, 0, 0), Rotation (0°, 0°, 0°), and Scale (1, 1, 1). Type those values explicitly if needed.
5. Press Esc or right-click to cancel an unfinished transform. Save only after you have restored the baseline.

WHY Axis constraints make movement predictable. A shared controller prevents the envelope, lines and basket from separating when you scale or rotate the assembly.

CHECK All four parts move together, and the controller is back at the baseline values before camera setup.

FIX Only one part moved? Undo and select BALLOON_CTRL. Never “fix” this by dragging the basket back by eye. Repair the hierarchy first.

Evidence: Explain why you used the controller rather than selecting Envelope alone.

Tool references: B4 • Full links on pp. 54–55.

Smooth the fabric, not the geometry

SELECT FIRST Envelope in Object Mode.



Actual classroom mesh rendered without the image texture; not a Blender interface capture.

- 1.** Choose Object > Shade Smooth, or press F3 over the viewport, search Shade Smooth and run it. This changes the shading of the existing faces.
- 2.** Inspect the silhouette and panel lines. Do not add Subdivision Surface, Remesh or a new UV unwrap. Those operations are unnecessary for the first lesson.
- 3.** Press Tab once to look at Edit Mode. Observe the many vertices and edges, but do not transform or delete them.
- 4.** Press Tab again to return to Object Mode. Check the mode label explicitly, then save.

WHY Object Mode moves complete objects. Edit Mode changes the mesh itself. Smooth shading changes how light is interpolated across faces; it does not add detail or repair an incorrect shape.

CHECK The fabric reads as a curved surface. You are back in Object Mode and the balloon has not changed shape.

FIX Everything is covered in selected orange edges? You are probably in Edit Mode. Press Tab. Unexpected dents require Undo, not additional smoothing.

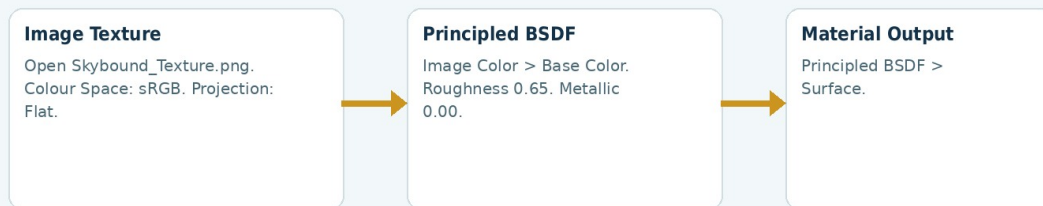
Tool references: B5 · Full links on pp. 54–55.

Open the material workspace

SELECT FIRST Envelope, with the model back in Object Mode.

07 / Connect colour — do not plug the image into Alpha

Teaching diagram • not a software screenshot



The existing UV map tells Blender where the image belongs.

Select Envelope first. A material on the basket will not decorate the balloon.

Material Preview shows the texture; Solid shading often does not.

Node names and connections are shown as a teaching diagram.

1. Click the Shading workspace tab. Find its 3D Viewport and Shader Editor. Confirm the Shader Editor context is Object rather than World.
2. Click the Material Preview sphere at the upper right of the viewport. Allow the preview shaders to load.
3. In Material Properties, confirm Envelope has MAT_Envelope. In the Shader Editor you should see Principled BSDF connected to Material Output.
4. If there is no node graph, enable Use Nodes for that material. If necessary, click New once to create a material for Envelope and name it MAT_Envelope.
5. Use the Shader Editor's View > Frame All to find nodes that are off screen. Do not disconnect the existing BSDF-to-Surface connection.

WHY Solid viewport shading is designed for inspecting shape and may not display image textures. Material Preview is useful for surface work, but its studio lighting is not necessarily your final scene lighting.

CHECK The selected object is Envelope and its material graph ends at Material Output > Surface.

FIX You see a Background node instead? Switch the Shader Editor from World to Object. Pink material later means a missing image, not a wrong fabric shape.

Tool references: B6 • Full links on pp. 54–55.

Connect the image to the material

SELECT FIRST Envelope's MAT_Envelope in the Shader Editor.

08 / A flat image wraps around the envelope

Teaching diagram • not a software screenshot



Keep the texture dimensions and existing UV map.



Flat artwork and an actual render using the supplied UV layout.

1. With the pointer in the Shader Editor, choose Add > Texture > Image Texture. Place the new node to the left of Principled BSDF.
2. Click Open on the Image Texture node. Choose starter/Skybound_Texture.png. Leave Colour Space at sRGB, Projection at Flat and interpolation at Linear.
3. Drag from the Image Texture node's yellow Color output to the Principled BSDF yellow Base Color input. Leave the Image Texture Alpha output unconnected.
4. Leave the Vector input unconnected: the image uses the existing active UV map. Keep Principled BSDF connected to Material Output > Surface.
5. Inspect the fabric in Material Preview. The large SKYBOUND lettering should wrap around the front. Save the .blend file.

WHY A colour swatch is not an image texture. This is the missing connection in the old material file. UV coordinates tell Blender where each part of the flat artwork belongs on the curved fabric.

CHECK Cream branding, teal panels and coral accents appear on Envelope, while the basket keeps its separate material.

FIX Grey fabric? Check Material Preview and the Color-to-Base Color connection. Transparent fabric? Remove any image connection to Alpha; the opaque fabric should not be cut out.

Evidence: Capture your own Shader Editor showing the image node and its connection.

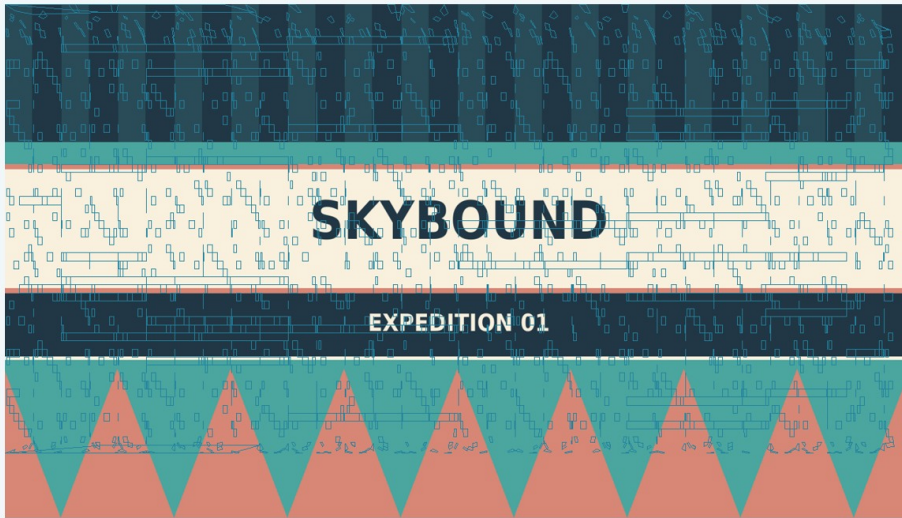
Tool references: B6 · Full links on pp. 54–55.

Inspect the existing UV map

SELECT FIRST Envelope only; do not select the basket.

09 / Inspect the UV layout — do not unwrap again

Teaching diagram • not a software screenshot



Blue mesh lines are a sample of the supplied UV layout, not a new projection.

Blue lines sample the actual model UV layout; fewer lines are shown for readability.

1. Switch to UV Editing. With the pointer over the 3D Viewport, press Tab for Edit Mode and A to select all Envelope faces.
2. In the UV Editor, choose Skybound_Texture.png from the image selector. Use View > Frame All to see the full layout.
3. Compare the UV edges with the flat bands and lettering. Some coordinates wrap around the image edge because the fabric wraps around a three-dimensional surface.
4. Do not choose Unwrap, Smart UV Project or Cube Projection. The supplied map is already aligned to this artwork.
5. Return to Object Mode with Tab and switch back to Layout or Shading. Save without changing the UV coordinates.

WHY A UV map is a flat address system for the mesh surface. Re-unwrapping a working model can scramble a correctly placed texture; inspection is the learning goal here.

CHECK The texture remains correctly wrapped after leaving UV Editing. You can explain where the flat artwork meets the curved model.

FIX No UVs visible? Check that Envelope is in Edit Mode with faces selected. The UV diagram is a reference; do not redraw its lines as new geometry.

Tool references: B6 • Full links on pp. 54–55.

Give the fabric a believable surface

SELECT FIRST Envelope's Principled BSDF node.

10 / Roughness describes the surface, not its colour

Teaching diagram • not a software screenshot



SHINY / HARD HIGHLIGHT



MATTE / SOFTER HIGHLIGHT

Conceptual shiny/matte comparison rendered from the model in VTK; not exact Blender Roughness outputs.

1. Set Metallic to 0.00 and Roughness to 0.65 on Principled BSDF. Keep Alpha at 1.00; do not add transmission or emission.
2. Briefly test Roughness 0.15. Observe the tighter, glossier highlight, then restore 0.65.
3. Leave the basket and rigging materials separate. A dark basket is acceptable at this stage; lighting is next.
4. Return to the front view and check the wordmark. Save the file.

WHY Roughness describes how concentrated a reflection is. The purpose is not to make every material shiny; a softer fabric highlight helps the image read as a balloon rather than polished plastic.

CHECK The fabric has a softer highlight, its colour remains visible, and the material is not metallic or transparent.

FIX The preview is still brighter than the final render? Material Preview can use a studio environment. Judge the final light with Rendered shading or an actual render, not Material Preview alone.

Tool references: B7 • Full links on pp. 54–55.

Add a fixed camera

SELECT FIRST Object Mode in the Layout workspace; 3D cursor at the world origin.

11 / A simple, reproducible camera

Teaching diagram • not a software screenshot



Camera_Render

Location: X 0, Y -26, Z 6. Rotation: X 90°, Y 0°, Z 0°. Lens Type: Orthographic. Scale: 14.4.

Output

1200 × 1200 px. 100% for final; 50% for tests. Pixel aspect: 1:1. Full balloon and basket inside the frame.

The camera uses the normalised, 12-unit classroom model.

1. Choose Add > Camera. Rename it Camera_Render using F2. Do not parent it to BALLOON_CTRL.
2. Open the Sidebar with N > Item. Set Location X 0, Y -26, Z 6. Set Rotation X 90°, Y 0°, Z 0°. Leave Scale at 1.
3. In Camera Data Properties (camera icon), change Type to Orthographic and set Orthographic Scale to 14.4.
4. Choose View > Cameras > Active Camera, or press Numpad 0. The camera looks from negative Y toward the front of the model.
5. The orthographic setup is a beginner framing aid. Keep it for the core; do not try to match the photograph by stretching the render.

WHY The camera, not your last orbit view, defines the rendered image. Orthographic projection gives us predictable framing for an isolated object and removes one variable from this first lesson.

CHECK You see the whole balloon and basket inside the camera view with comfortable space around them.

FIX Camera points away? Re-enter Rotation (90°, 0°, 0°). Model is not upright? Revisit the import axes rather than rotating the camera to compensate.

Tool references: B8 • Full links on pp. 54–55.

Set a consistent render canvas

SELECT FIRST Output Properties in the Properties editor.

11 / A simple, reproducible camera

Teaching diagram • not a software screenshot



Camera_Render

Location: X 0, Y -26, Z 6. Rotation: X 90°, Y 0°, Z 0°. Lens Type: Orthographic. Scale: 14.4.

Output

1200 × 1200 px. 100% for final; 50% for tests. Pixel aspect: 1:1. Full balloon and basket inside the frame.

The guide's prepared PNGs use the same 1200 × 1200 canvas, but a different render engine.

1. Under Format, set Resolution X to 1200 px, Y to 1200 px, and Percentage to 100%. Set Pixel Aspect X and Y to 1.000.
2. Check the camera view again. The square frame must include the entire basket and fabric. Leave the camera at the numeric setup from Step 13.
3. In Render Properties > Color Management, use the AgX View Transform and leave the other display settings at their defaults for the first tests. Do not change the view transform between A and B renders.
4. For a faster test later, change only Percentage to 50%. Return it to 100% before saving final renders.
5. Save the .blend file. A consistent canvas will make replacing a render in Photoshop much easier.

WHY 1200 × 1200 is a pixel dimension, not a print-size instruction. Keeping the same transparent margins in each render lets an updated angle replace the old PNG without an unexpected size jump.

CHECK The camera frame is square and the entire balloon fits. Resolution remains 1200 × 1200 even when a test percentage is lower.

FIX Your render is 600 × 600? Percentage is still 50%. A close-up is not fixed by increasing resolution; framing belongs to the camera.

Tool references: B8, B9 • Full links on pp. 54–55.

Light the balloon from the scene's left

SELECT FIRST Object Mode; no existing light selected.

12 / Key light first, gentle fill second

Teaching diagram • not a software screenshot

KEY — warm, upper left

Area / 1800 W / Size 5. Position (-6, -8, 13). Rotation (59.04°, 0°, -36.87°).



FILL — weaker, right

Area / 500 W / Size 6. Position (5, -4, 7). Rotation (90°, 0°, 51.34°).

Starting values for the supplied 12-unit model. Judge the rendered result.

Area-light positions and rotations are lesson setup values; the diagram is not a Blender screenshot.

1. Choose Add > Light > Area. Rename the light Key_Area. Open N > Item for its transforms.
2. Set Location to (-6, -8, 13). Set Rotation to (59.04°, 0°, -36.87°). Do not rotate the balloon instead.
3. In Light Data Properties (lightbulb icon), set Power to 1800 W, Shape to Square and Size to 5. Set the light colour to a pale warm tint, such as hex FFE5C5.
4. In Render Properties choose Eevee. Switch the viewport to Rendered shading, then look through Camera_Render.
5. Observe the envelope's upper-left side. The key should create the dominant light direction; the values are starting points for this prepared model, not universal settings.

WHY The photograph already has a strong left-side light. Matching that direction in 3D is more convincing than painting a contradictory bright edge in Photoshop later.

CHECK The left/top fabric is more illuminated than the right side. The light points toward the balloon rather than away from it.

FIX Black result? Check Power, Rendered mode and the light's rotation. A flat white patch means the light may be too strong; halve its power and test before changing all settings.

Tool references: B10 • Full links on pp. 54–55.

Add gentle fill and ambient light

SELECT FIRST A new Area light; keep Key_Area unchanged.

12 / Key light first, gentle fill second

Teaching diagram • not a software screenshot

KEY — warm, upper left

Area / 1800 W / Size 5. Position (-6, -8, 13). Rotation (59.04°, 0°, -36.87°).



FILL — weaker, right

Area / 500 W / Size 6. Position (5, -4, 7). Rotation (90°, 0°, 51.34°).

Starting values for the supplied 12-unit model. Judge the rendered result.

EEVEE and Cycles can respond differently to World illumination; use test renders to judge the result.

1. Choose Add > Light > Area and rename it Fill_Area. Set Location to (5, -4, 7) and Rotation to (90°, 0°, 51.34°).
2. In Light Data Properties set Power to 500 W, Shape to Square, Size to 6 and colour to neutral white. The fill must remain weaker than the key.
3. Open World Properties > Surface. Use Background, set Color to a medium neutral grey (hex 808080 as a starting point), and set Strength to 0.30.
4. Inspect Camera_Render in Rendered mode. If EEVEE does not give you useful World fill without additional probes, use the Area fill as the core control rather than building a probe system.
5. Check the Outliner: the camera and two lights must be beside BALLOON_CTRL, not underneath it. Save.

WHY Fill light preserves detail on the darker side without creating a competing sun. World lighting and preview lighting behave differently across render engines, so the Area lights are the predictable core setup.

CHECK You can see the basket and right-side fabric, but the left side still reads as the dominant light source.

FIX Turning the balloon moves the lights? They were accidentally parented. Select the misplaced lights and use Object > Parent > Clear Parent (Keep Transform).

Tool references: B10 • Full links on pp. 54–55.

Test one lighting decision at a time

SELECT FIRST Camera_Render view, Rendered shading.

14 / Prove that you changed the real 3D model

Teaching diagram • not a software screenshot



A: CTRL rotation Z = 0°



B: CTRL rotation Z = 30°

These source-model supports show useful form and texture visibility, not a required pixel-for-pixel Blender match.

1. Compare your balloon with the left-side light in Landscape.jpg. Look at the fabric, not just the background colour of the viewport.
2. Test Key_Area at 900 W, then at 1800 W. Return to the value that preserves the cream band without losing the darker side. Write down your chosen value.
3. Check Fill_Area separately. Keep it weaker than the key; adjust only its Power when the basket or right-side fabric is too dark.
4. Keep the texture and camera unchanged during this test. Do not use emission to make the label brighter.
5. When the direction and contrast are clear, save the .blend. Your render need not have the identical colour of the VTK teaching example.

WHY A controlled comparison tells you which setting caused the change. Changing light power, material colour, camera and grading at once makes a problem hard to diagnose.

CHECK The branding remains readable, the envelope is visibly curved, and the image has one main light direction.

FIX Material Preview looks good but a render is dark? Material Preview may be using its own studio lights. Test your actual scene lights in Rendered mode or with Render > Render Image.

Evidence: Record one lighting change and the visible effect it produced.

Tool references: B10 · Full links on pp. 54–55.

Set up a truly transparent render

SELECT FIRST Render Properties, then Output Properties.

13 / PNG needs alpha — transparency is not a black background

Teaching diagram • not a software screenshot



Two places to check

Render Properties > Film > Transparent: ON.
When saving: PNG / RGBA / 8-bit / Save As
Render ON.

A means Alpha

RGB saves colour only. RGBA also saves transparency. Test the PNG over two different background colours.

The supplied PNG is displayed over a checkerboard to verify its actual alpha channel.

1. In Render Properties expand Film and enable Transparent. This hides the background from the camera render; it does not make the balloon transparent.
2. In Output Properties choose PNG as the image format, RGBA as the colour channels, and 8-bit colour depth for this classroom composite.
3. Use Output Percentage 50% for the first test. Leave Camera_Render active, the model visible and both Area lights enabled.
4. Open Render > Render Image, or press F12. A checkerboard should surround the object in the Render Result.
5. Remember that these scene settings do not replace the separate Image > Save As step. The save dialog must also retain PNG and RGBA.

WHY RGB stores colour; RGBA adds an alpha channel for transparency. A JPEG cannot carry this transparent background, and a black viewer background is not proof that alpha is missing.

CHECK The rendered balloon is opaque, but the empty space and gaps between the ropes show a checkerboard.

FIX Solid background? Check Film > Transparent, then rerender. Transparent fabric? Check the material Alpha is 1 and the image Alpha is not connected to it.

Tool references: B11, B12 · Full links on pp. 54–55.

Save and inspect the small test render

SELECT FIRST The Render Result in Blender's Image Editor.

13 / PNG needs alpha — transparency is not a black background

Teaching diagram • not a software screenshot



Two places to check

Render Properties > Film > Transparent: ON.
When saving: PNG / RGBA / 8-bit / Save As
Render ON.

A means Alpha

RGB saves colour only. RGBA also saves transparency. Test the PNG over two different background colours.

Test the saved file's alpha, not just its appearance in a viewer.

1. Choose Image > Save As from the Render Result window. Navigate to My_Renders and name the file Balloon_TEST.png.
2. In the save options confirm PNG, RGBA, 8-bit and Save As Render enabled. Click Save As Image and confirm the location if prompted.
3. Open the saved PNG to check the real exported file, not just the Render Result. It should be 600 × 600 when the 1200-pixel scene was rendered at 50%.
4. Inspect the basket, wordmark and thin ropes. A black or white background in an image viewer can simply be its way of displaying transparency; test over another colour when uncertain.
5. Return to Blender and correct one issue before the final renders. Do not enlarge this small test to create your final image.

WHY Saving a .blend stores the scene, not a guaranteed standalone image export. A test render finds framing, texture and alpha problems before you spend time on the final.

CHECK Balloon_TEST.png exists in My_Renders and contains the whole model, correct texture and transparency.

FIX The file is absent? F12 rendered it but did not save it. Use Image > Save As. A .blend file renamed .png is not an image export.

Evidence: Keep the test file and note one improvement for the final render.

Tool references: B12 · Full links on pp. 54–55.

Render two genuinely different 3D views

SELECT FIRST BALLOON_CTRL; then the Render Result after each render.

14 / Prove that you changed the real 3D model

Teaching diagram • not a software screenshot



A: CTRL rotation Z = 0°



B: CTRL rotation Z = 30°

Actual geometry rotated between renders. Students must create their own A/B files for 3D evidence.

1. Restore Output Percentage to 100%. Set BALLOON_CTRL Rotation to (0°, 0°, 0°). Keep Camera_Render, lights and resolution unchanged.
2. Choose Render > Render Image. Save through Image > Save As as My_Renders/Balloon_A_Front.png, PNG/RGBA/8-bit with Save As Render on.
3. Select BALLOON_CTRL again and set only Rotation Z to 30°. Render again and save Balloon_B_Turned.png with the same image settings.
4. Open both PNGs side by side. Look for changed lettering position and a different visible side, not merely a flat picture tilted clockwise.
5. Return the controller to Z = 0° for the core composition and save the .blend. Keep both renders as evidence of your 3D work.

WHY Rotating the model before rendering changes what the camera sees around its curved surface. Rotating a PNG in Photoshop cannot reveal another side of the balloon.

CHECK Both files are 1200 × 1200 with transparency. The model changes angle while the camera and light direction remain fixed.

FIX Only the envelope turned? Fix parenting. Both images look identical? Check that you changed BALLOON_CTRL, rerendered, and saved the new Render Result under a different name.

Evidence: Submit your own A and B renders, plus a sentence comparing them.

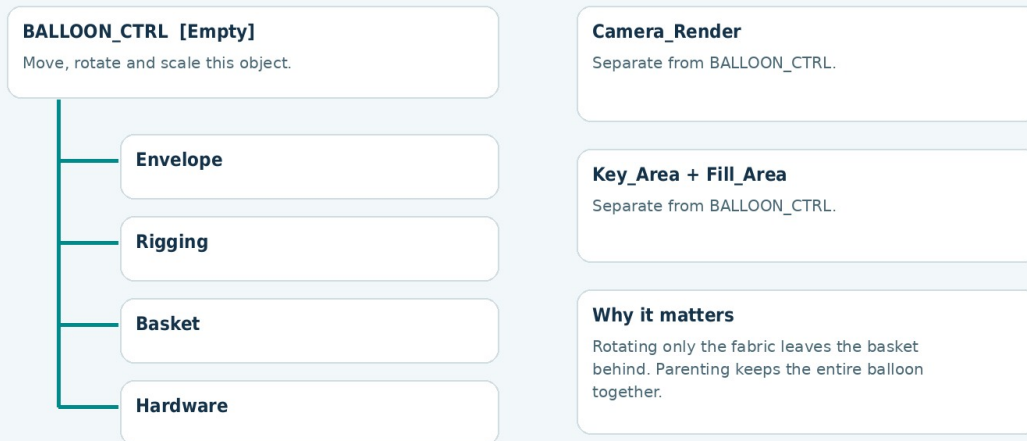
Tool references: B4, B12 · Full links on pp. 54–55.

Save a portable editable scene

SELECT FIRST The completed scene in the Layout workspace.

06 / Parent the parts — keep the camera independent

Teaching diagram • not a software screenshot



Your .blend is student-created; the resource pack does not contain a pretested Blender scene.

1. Choose File > External Data > Pack Resources. Then save the .blend file again; the packing is written into the saved scene.
2. Confirm the file name is 3D_Work/Lastname_Skybound.blend. Do not overwrite the classroom OBJ or your texture artwork.
3. Close and reopen your saved .blend after saving all work. Inspect the material and hierarchy to check that the file opens correctly.
4. Keep the original source folder and My_Renders as well. Packed textures are helpful, but they do not replace your separate PNG exports.
5. Capture your own Outliner/scene view showing BALLOON_CTRL, its four children, Camera_Render, Key_Area and Fill_Area.

WHY A portable scene preserves the relationship between the model, materials, lights and camera. A rendered PNG is only a view of that scene, not a replacement for the editable project.

CHECK The reopened scene still shows its artwork, and your two separate final PNGs remain available.

FIX Pink texture after reopening? Locate the image in the Image Texture node, reopen it, then Pack Resources and save again. Do not assume an unsaved Pack command is enough.

Evidence: Submit a scene screenshot and the packed .blend file.

Tool references: B13 · Full links on pp. 54–55.

Start the Photoshop composition

SELECT FIRST File > New; a fresh RGB document.

15 / Place the complete render without stretching

Teaching diagram • not a software screenshot



Layout target

2000 × 1333 canvas.
Balloon PNG 1200 × 1200. Centre X 1430 / Y 650. Leave W and H linked.

The layout diagram uses the landscape's native 2000 × 1333 pixel canvas.

1. Choose File > New. Set Width 2000 pixels, Height 1333 pixels, Resolution 72 pixels/inch, RGB Color, 8-bit and sRGB IEC61966-2.1. Use a white background.
2. Name the document Lastname_Skybound. These dimensions match the supplied landscape; resolution here is for a digital composition, not a large-print claim.
3. Open Window > Layers and Window > Properties. Use View > Rulers if guides help you plan the layout.
4. Save as Photoshop_Work/Lastname_Skybound.psd. Keep layers enabled. The file is initially almost empty; that is expected.
5. Use the core reference to reserve the right side for the main balloon and the open left side for depth or later typography.

WHY A shared canvas makes position, brush-size and output instructions meaningful. Matching the landscape's native dimensions avoids unnecessary enlargement of that photograph.

CHECK The document is 2000 × 1333 pixels, RGB/8, with the Layers panel visible.

FIX Dimensions entered in centimetres? Switch both units to pixels before creating. A 72-ppi label does not mean the image has only 72 pixels of detail.

Tool references: P1 • Full links on pp. 54–55.

Place the landscape without flattening it

SELECT FIRST The new composition document.



The original landscape with a restrained saturation reduction; no AI replacement background.

1. Choose File > Place Embedded and select starter/Landscape.jpg. Fit the entire photograph to the canvas without cropping or stretching. Press Enter/Return to commit.
2. Rename the layer LANDSCAPE. Confirm the thumbnail has a Smart Object badge. Leave its blend mode Normal and opacity 100%.
3. Choose Layer > New Adjustment Layer > Hue/Saturation. Name it LANDSCAPE_SAT and click OK.
4. With that adjustment selected, choose Layer > Create Clipping Mask so it affects only LANDSCAPE. Set Master Saturation to -16 as a starting point; leave Hue and Lightness at 0.
5. Delete the unused white Background layer only after the photograph covers the entire canvas. Save.

WHY An embedded Smart Object preserves the placed source. Reducing background saturation slightly helps the new balloon stand out without changing the photograph into a different scene.

CHECK The full landscape fills the canvas, and LANDSCAPE_SAT has a clipping indicator pointing to LANDSCAPE.

FIX Adjustment affects a later balloon too? It is not clipped correctly. Use Layer > Create Clipping Mask while the adjustment is directly above LANDSCAPE.

Tool references: P1, P2 · Full links on pp. 54–55.

Place your transparent balloon render

SELECT FIRST LANDSCAPE_SAT, so the new layer is placed above the background correction.



Source-model PNG placed into the supplied landscape. Your own Blender shading may differ.

1. Choose File > Place Embedded and select your My_Renders/Balloon_A_Front.png. Do not use the supplied support unless your teacher has approved the supported route.
2. Rename the new Smart Object BALLOON_RENDER. Keep the proportions linked and the blend mode Normal at 100% opacity.
3. For a complete 1200 × 1200 PNG canvas, the target centre is X 1430, Y 650. In Free Transform, use the centre reference point and set W/H to 100% when Photoshop imported the image at its original size.
4. When the version trims transparent margins, use the visual reference instead: place the visible envelope in the upper-right area and the basket above the foreground, with all parts on canvas. Do not stretch it to force the numeric target.
5. Commit with Enter/Return. Confirm the landscape is visible through rope gaps and around the object. Save.

WHY Transparency carries the object's silhouette. A Smart Object preserves the rendered source, but it remains a flat PNG inside Photoshop; it is not a live imported 3D model.

CHECK The main balloon occupies the right third, the full basket is visible, and there is no white or black rectangle.

FIX Opaque rectangle? Return to the Blender PNG/RGBA export; do not erase it by hand. Tiny result? Check whether Photoshop scaled the placed file to fit and use the visual reference.

Evidence: Record whether you used your own render or an approved prepared support.

Tool references: P1 · Full links on pp. 54–55.

Match tone with a clipped adjustment

SELECT FIRST BALLOON_RENDER in the Layers panel.



A teaching composite showing restrained local matching, not a pixel-exact Photoshop Curves result.

1. Choose Layer > New Adjustment Layer > Curves. Name it BALLOON_TONE. Place it immediately above BALLOON_RENDER and choose Layer > Create Clipping Mask.
2. In Properties, add an RGB curve point near the middle. Use Input 128, Output 120 as a mild starting darkening; leave the end points unchanged.
3. Toggle the adjustment's eye icon to compare. Keep readable highlights on the cream label and visible detail in the basket.
4. Your Blender lighting may already match well. Reduce the correction or leave the curve close to neutral when a darker result loses the form.
5. Inspect the balloon against nearby sky, not only against bright highlights. Save the file.

WHY Local tone matching brings the render into the photograph's exposure range. The numbers are a test, not a rule: a correction that crushes the subject is not an improvement.

CHECK Only the balloon changes when you toggle BALLOON_TONE. The main highlight stays on the same side as the scene light.

FIX The whole photograph darkened? Clip the Curves layer to BALLOON_RENDER. If the balloon's light is on the wrong side, fix it in Blender rather than trying to reverse it with one curve.

Tool references: P2 · Full links on pp. 54–55.

Guide a subtle highlight with a mask

SELECT FIRST BALLOON_TONE; the next adjustment must remain in the balloon's clipping stack.

16 / Clip the correction to the balloon

Teaching diagram • not a software screenshot

BALLOON_LIGHT

Curves • clipped • black mask

BALLOON_TONE

Curves • clipped

BALLOON_RENDER

Embedded Smart Object

LANDSCAPE_SAT

Hue/Saturation • clipped

LANDSCAPE

Embedded Smart Object

Clipped adjustments preserve the landscape and let you revise local light independently.

1. Add another Curves adjustment named BALLOON_LIGHT above BALLOON_TONE. Use Layer > Create Clipping Mask so it also clips to BALLOON_RENDER.
2. Set a gentle brightening point, Input 128 and Output 144. Click the adjustment's white mask thumbnail, then press Ctrl+I (Windows) or Cmd+I (Mac) to make the mask black.
3. Press B for Brush. Choose a soft round brush, Hardness 0%, Size about 250 px, Opacity 15%, Flow 100%. Set the foreground colour to white.
4. Paint a few light strokes on the mask over the envelope's upper-left fabric. Keep the wordmark readable and the right side darker.
5. Toggle BALLOON_LIGHT to compare. Lower the adjustment opacity or paint black on its mask to reduce an exaggerated highlight.

WHY A black mask hides the correction; white paint reveals it selectively. You are reinforcing the existing light, not adding a new white stripe or pretending the material emits light.

CHECK The highlight is restrained, curved and restricted to the balloon. You see a soft white patch on the black adjustment mask.

FIX White paint appears on the picture? You selected an image layer, not the mask. Undo, click BALLOON_LIGHT's mask thumbnail and try again.

Evidence: Capture one mask and explain what white and black mean in that mask.

Tool references: P2 • Full links on pp. 54–55.

Add depth without hiding your work

SELECT FIRST A new normal pixel layer above the balloon's adjustment stack.

17 / Match light, then add just enough atmosphere

Teaching diagram • not a software screenshot



BEFORE MATCHING



AFTER MATCHING + MIST

Keep fabric detail. Let a little mist cross the lower basket and rigging.

The reference mist is a prepared soft-painted layer, not a photographic cloud cut-out.

1. Choose Layer > New > Layer. Name it FOG_FRONT. Do not clip this layer: atmosphere should be able to cross both the object and background.
2. Set the foreground colour to pale blue-grey, hex CBDBD8. Use a soft round Brush, Hardness 0%, Size 400–700 px, Opacity 10%, Flow 100%.
3. Paint a small amount near the lower basket and rigging, then a faint horizontal patch in the distant valley. Leave the upper branding and most of the envelope clear.
4. Lower the layer opacity when the fog looks like a solid brush blob. Erase nothing from the balloon; use a layer mask on FOG_FRONT to hide excess fog.
5. An approved alternative is File > Place Embedded > Mist_Support.png above the balloon at full canvas size. Use either your painting or the support first, not both at full strength.

WHY An overlapping foreground layer helps the object sit within an atmosphere rather than on top of the photograph. A floating balloon does not need a tight contact shadow immediately under its basket.

CHECK Only a small area of the lower balloon overlaps mist. The ropes, basket and wordmark are still readable.

FIX The balloon looks transparent everywhere? Restore BALLOON_RENDER to 100% opacity. Atmospheric fading should be a separate local effect, not see-through fabric.

Tool references: P2 • Full links on pp. 54–55.

Unify the scene with a restrained grade

SELECT FIRST The top of the entire layer stack, above FOG_FRONT.

18 / Final core layer order

Teaching diagram • not a software screenshot

GLOBAL_WARMTH	Photo Filter • affects the scene
FOG_FRONT	Painted pixel layer • not clipped
BALLOON_LIGHT	Curves • clipped to balloon
BALLOON_TONE	Curves • clipped to balloon
BALLOON_RENDER	Smart Object • your own PNG
LANDSCAPE_SAT	Hue/Saturation • clipped to landscape
LANDSCAPE	Smart Object • supplied photograph

Core layer order; arrows represent clipping relationships, not extra image layers.

1. Choose Layer > New Adjustment Layer > Photo Filter. Name it GLOBAL_WARMTH and leave it unclipped so it affects the whole scene.
2. Select Warming Filter (85), set Density to 8%, and enable Preserve Luminosity. Treat this as a starting point.
3. Compare with the adjustment off. Reduce its Density when the cream label becomes yellow or the greens lose their character.
4. Check the final order against the layer diagram: GLOBAL_WARMTH, FOG_FRONT, the two clipped balloon adjustments, BALLOON_RENDER, clipped LANDSCAPE_SAT, LANDSCAPE.
5. Save the layered PSD. The core image is now complete enough for the round-trip and quality checks.

WHY A small common colour bias can connect sources. Strong colour grading is not a substitute for matching perspective, transparency or light direction.

CHECK The image feels unified without an orange wash, muddy shadows or unreadable lettering.

FIX The grade affects only the balloon? It was clipped accidentally. Release its clipping mask and keep it at the top. The local corrections below should stay clipped.

Tool references: P2 • Full links on pp. 54–55.

Round-trip one revision through Blender

SELECT FIRST Your saved .blend; then BALLOON_RENDER in Photoshop.

01 / One object. Two creative spaces.

Teaching diagram • not a software screenshot

BLENDER

Import your model. Apply its artwork. Choose a camera and light. Render a transparent image.



PNG + ALPHA

A flat image of your 3D object. Empty pixels remain transparent. This is the bridge between apps.



PHOTOSHOP

Place the render as a Smart Object. Match its tone. Add subtle atmosphere. Export the composite.

MODEL ≠ MATERIAL ≠ RENDER

Geometry gives the shape. A texture supplies colour. Light reveals the form.

Change the 3D camera or material in Blender, then render again.

This is a render-replacement workflow, not the discontinued live Photoshop 3D workflow.

1. In Blender, make one small deliberate change: for example, change BALLOON_CTRL Z rotation from 0° to 10°, or lower Key_Area power slightly.
2. Keep the same camera, 1200 × 1200 resolution and transparent margins. Render a new image and save Balloon_C_Revision.png as PNG/RGBA.
3. In Photoshop, select BALLOON_RENDER. Choose Layer > Smart Objects > Replace Contents and select Balloon_C_Revision.png.
4. Inspect the updated image. The transformation should be retained, but revise local highlight and fog masks when the new angle changes where they belong.
5. Keep the version you can justify. Undo the replacement if A communicates better, but retain C and your explanation as revision evidence.

WHY A Photoshop Smart Object containing a PNG does not contain the editable 3D scene. To change the model's viewpoint or lighting, return to Blender and replace the rendered contents.

CHECK A real 3D revision appears in the PSD, while the original layered structure stays editable.

FIX Double-clicking the Smart Object only opened a PNG? That is expected. A sudden scale jump often means the new render had different dimensions or was cropped differently.

Evidence: Write: "I changed ___ in Blender. In the composite, that improved/changed ___."

Tool references: P1 · Full links on pp. 54–55.

Inspect at 100%, then at thumbnail size

SELECT FIRST The full composition, with all intended core layers visible.

19 / Four checks before you submit

Teaching diagram • not a software screenshot



1 Readable label – 2 Alpha in thin ropes – 3 Light direction – 4 Depth / atmosphere

Four inspection targets: branding, thin geometry, light direction and depth.

1. Choose View > 100%. Pan around the envelope edge, rope gaps and basket. Look for a background rectangle, dark fringe, clipped parts or accidental blur.
2. Check that the label is readable and highlights are not blown out. Compare the light direction on the balloon with the upper-left light in the landscape.
3. Choose View > Fit on Screen, then zoom farther out. The main balloon should be the first thing you notice, with the landscape supporting its scale.
4. Toggle FOG_FRONT and the local adjustments individually. Keep only changes that help the image. Use the correct mask or 3D source to repair the cause.
5. Save a final layered version before trying any optional extensions. Do not flatten the only editable file.

WHY Close inspection catches technical problems; a thumbnail test catches composition problems. Both matter, and one cannot replace the other.

CHECK The full silhouette is intact, alpha works, the light agrees, and the subject is clear at small viewing size.

FIX Over-sharpened outline? Return to the clean render rather than blurring the whole composition. Do not use a heavy vignette to hide unfinished edges.

Tool references: P1, P2 · Full links on pp. 54–55.

Save the editable work and export the artwork

SELECT FIRST Your final core PSD and saved .blend.

02 / Know the files before you start

Teaching diagram • not a software screenshot

Balloon_Classroom.obj

Shape + UV coordinates

Balloon_Classroom.mtl

Four starting material colours

Skybound_Texture.png

Artwork for the curved envelope

Landscape.jpg

The 2D scene behind the balloon

Balloon_A_Front.png

Optional prepared render support

Student_Skybound.blend

Your editable 3D scene — you create it

Your project folder should contain sources, editable work, evidence and exports.

1. Save Photoshop_Work/Lastname_Skybound.psd with layers. Also save 3D_Work/Lastname_Skybound.blend with packed resources.
2. In Photoshop choose File > Export > Export As. Select JPG, keep Width 2000 and Height 1333, and enable Convert to sRGB and Embed Color Profile when available.
3. Use a high quality setting, around 85% when the dialog uses a 0–100 scale. Export as Lastname_Skybound_Final.jpg. Do not change the PSD's only copy to JPEG.
4. Open the exported JPEG and check it. Confirm it is the intended version, not a test render or a screenshot of the workspace.
5. Keep your A, B and C transparent PNGs in My_Renders. Export or back up your guide notes too; browser storage alone is not a submission.

WHY Each file has a different job: .blend for editable 3D, .psd for the layered composition, PNG for transparent renders and JPG for the finished flat image.

CHECK All four file types are present where expected. The final JPEG is 2000 × 1333 and matches your chosen composition.

FIX Only a JPEG exists? Reopen the layered working file; a JPEG cannot recover the original layers. A render is not automatically saved when you save the .blend.

Tool references: B12, B13, P3 • Full links on pp. 54–55.

Explain the decisions, not just the clicks

SELECT FIRST Your final JPEG, .blend, PSD and your own A/B render pair.

SKYBOUND / Core first. Storytelling second.

Source-model teaching composites • no AI-generated replacement model



CORE / one 3D balloon



EXTENSION / a campaign world

The optional effects do not replace evidence of importing, texturing and rendering.

Core success is a controlled 3D-to-2D workflow; extra effects are optional.

1. Submit the final JPEG, layered PSD, packed .blend, two different-angle PNGs, and your exported process notes using your teacher's submission method.
2. Include a screenshot of the image-texture node connection and one of the parent hierarchy. Use your own screenshots; guide diagrams are not evidence of your work.
3. Answer: What changed when you rotated the model rather than the PNG? How did the UV map place the wordmark? Why did you need RGBA?
4. Explain one lighting decision and one compositing revision. Point to a visible change, not only a setting number.
5. Declare any prepared renders, mist or other supports used. Record source-file credits and the permission status before public posting.

WHY A convincing image can hide a weak understanding of the process. The scene, render comparison and short explanations make your independent learning visible.

CHECK Your evidence demonstrates importing, texturing, changing a real 3D angle, saving transparency and integrating the result.

FIX Used the supplied render only? Submit the compositing work honestly and arrange a separate 3D demonstration. Do not claim the support as your own render.

Evidence: Complete the reflection and support declaration before checking this step off.

Design your own expedition identity

SELECT FIRST starter/Texture_Blank.png opened as a separate document.

08 / A flat image wraps around the envelope

Teaching diagram • not a software screenshot



Keep the texture dimensions and existing UV map.



The prepared texture is a starting layout, not a mandatory brand identity.

1. Open Texture_Blank.png. Preserve its 1571 × 887 dimensions and existing bands. Save a layered copy as Photoshop_Work/My_Expedition_Texture.psd.
2. Add a centred text layer in the cream band. Use a bold installed sans-serif font, about 80–95 px, dark navy 173B52. Keep the main word within roughly X 430–1140 and Y 315–435.
3. Add a short secondary line in the dark middle band near Y 535. Start around 36–42 px, warm cream FFF0D2. Use an original fictional expedition name; avoid unverified real-event claims.
4. Keep important lettering away from the far left and right edges where the wrap seam falls. Do not paint shadows into the artwork; the 3D lights will create them.
5. Export a full-size PNG named My_Expedition_Texture.png. Keep the layered PSD so you can change the text after seeing it on the balloon.

WHY A texture is a design surface, not a finished flat poster. Its layout must survive curvature and a seam. Short, bold branding is a practical first test.

CHECK Your PNG has the same dimensions as the supplied texture and preserves the band positions.

FIX Long words wrap around the sides? Shorten the name before shrinking it into unreadability. Keep a wider safe margin around the central label.

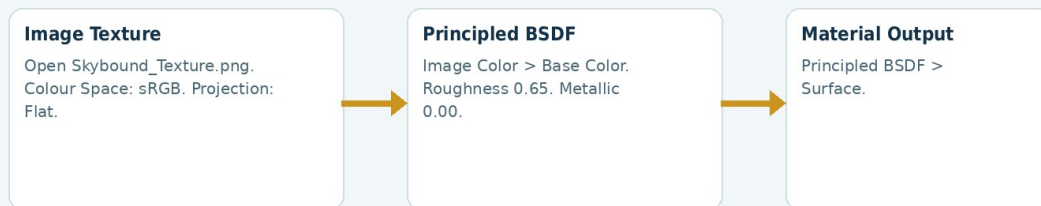
Tool references: P3, B6 · Full links on pp. 54–55.

Apply the new artwork and rerender

SELECT FIRST Envelope's Image Texture node, then the Photoshop Smart Object.

07 / Connect colour — do not plug the image into Alpha

Teaching diagram • not a software screenshot



The existing UV map tells Blender where the image belongs.

Select Envelope first. A material on the basket will not decorate the balloon.

Material Preview shows the texture; Solid shading often does not.

Use the original UV layout; the image changes, not the model's unwrap.

1. In Blender open the Image Texture node's image selector or Open control and load My_Expedition_Texture.png. Keep its Color output connected to Base Color.
2. Inspect the front and a turned view in Material Preview. Correct text position in the texture PSD when the curve hides important letters; export the PNG again.
3. When reusing the same image file name, reload it through the Image Editor's Image > Reload after saving the new PNG. Packed copies may need reopening from the updated external file.
4. Render with the same camera and transparent canvas. Save Balloon_Custom.png, then use Photoshop > Layer > Smart Objects > Replace Contents on BALLOON_RENDER.
5. Review the local masks, save the custom PSD, and Pack Resources plus save the updated .blend again.

WHY The revision loop connects graphic design to a 3D surface. It also teaches why editing a texture file does not necessarily update an already rendered PNG.

CHECK The new fictional identity appears on the curved fabric in the rendered image, not just as flat text pasted over the composite.

FIX Old text persists? Reload the correct image in Blender and rerender. Updating the texture cannot retroactively change an old PNG.

Tool references: B6, B13, P1 · Full links on pp. 54–55.

Build a small balloon fleet

SELECT FIRST A saved copy of the core PSD; work below the main balloon and fog.



Prepared fleet composite uses smaller real model renders and atmospheric colour matching.

1. Choose File > Save As and create Lastname_Skybound_Fleet.psd. Keep the completed core version unchanged.
2. Place your own Balloon_B_Turned.png twice as separate Smart Objects below the main balloon. Start with one complete PNG about 390 × 390 at centre (825, 375), and another about 225 × 225 at centre (513, 473).
3. Keep each balloon at 100% layer opacity. Reduce its contrast with a clipped Curves adjustment and add a small pale blue-grey clipped colour overlay at about 15–30% opacity to suggest haze.
4. Use the smaller, hazier balloon for greater distance. Keep all baskets clear of awkward overlaps and do not make every balloon the same angle.
5. Compare at thumbnail size. The large right balloon must remain the focus. Save the extension separately.

WHY Scale, contrast and overlap establish distance. Lowering the whole balloon layer's opacity makes solid fabric see-through; mixing its colour toward the atmosphere is a more useful depth cue.

CHECK The two smaller balloons feel farther away, but the background does not show straight through their fabric.

FIX The fleet competes with the main subject? Reduce the smaller balloons' scale or contrast, not the main balloon's readability.

Tool references: P1, P2 · Full links on pp. 54–55.

Add a deliberately surreal mirage city

SELECT FIRST A separate extension copy; this effect is not required for the core.



A separate mirage-city variation; optional fantasy, not the expected beginner outcome.

1. Place starter/CityOutline.png below the main balloon and foreground mist. Scale proportionally to about 240 px wide and place it in the far valley, around top-left (870, 410).
2. Use clipped Hue/Saturation with Colorize as a starting point: Hue about 155, Saturation about 20, then adjust Lightness slightly upward. Keep the city pale and low-contrast.
3. Add a layer mask to the city. Paint black with a soft 60–100 px brush over the lower building edges so the base disappears gradually into distant haze.
4. For an intentional apparition, set the city opacity near 50–60%. This transparency is a fantasy choice, unlike the solid balloons in E3. Name the layer `MIRAGE_CITY`.
5. Keep the effect small. Save this as a separate story version and explain why the city is a mirage rather than a physically solid settlement.

WHY An explicitly surreal effect can use transparency intentionally. Naming the visual rule prevents a accidental ghost-like composite from being mistaken for realistic atmospheric perspective.

CHECK The city's base fades naturally; there is no rectangular cut-out or hard horizontal building base. It does not cover the main wordmark.

FIX The city looks pasted on? Refine its base mask and lower contrast. Do not rely only on a glow to disguise a bad edge.

Tool references: P2 · Full links on pp. 54–55.

Frame the scene with controlled storm clouds

SELECT FIRST A new extension copy; use Clouds_Storm.jpg, not the unfinished original PSD.



An optional masked-cloud composition made from the supplied storm photograph.

1. Place starter/Clouds_Storm.jpg above the landscape but below all balloons. Scale it across the upper canvas. Keep blend mode Normal initially.
2. Add a layer mask and invert it to black with Ctrl/Cmd+I. Choose a soft white Brush at 15% opacity and 300–500 px size.
3. Reveal clouds near the top corners only. Leave a broad opening around the original upper-left light and preserve the horizon and rock silhouettes.
4. Lower the cloud layer opacity to about 30–50% as a starting point. Add a clipped Hue/Saturation correction when the cloud tint conflicts with the scene.
5. Judge whether the new cloud framing strengthens the story. Keep the clean core version too. A tornado is deliberately outside this first-3D lesson.

WHY Selective storm framing adds mood without hiding the 3D work or inventing a contradictory light source. A layer mask offers much more control than placing a cloudy rectangle over everything.

CHECK There is an open light path, a readable balloon and no straight cloud-photo edge across the sky.

FIX The entire image becomes dark and busy? Paint black on the mask and remove most of the effect. Keep the top corners, not a full-screen storm overlay.

Tool references: P2 · Full links on pp. 54–55.

Turn the scene into an expedition poster

SELECT FIRST A new saved copy of the core or fleet PSD; choose a fictional campaign.



Source-based campaign exemplar with live-type construction instructions; not a real travel offer.

1. Add a blank layer beneath new type but above the scene. Name it TYPE_BACKING. With a very large soft navy brush at low opacity, darken only the lower-left area; leave the main balloon untouched.
2. Use the Type tool to add SKYBOUND or your fictional expedition name near X 95, Y 1030. Start with a bold installed sans-serif at about 110–120 px, colour FFF0D2. Keep this as live type.
3. Add a smaller line near X 100, Y 1180, around 30–34 px: “A new perspective starts here.” Add a small “Fictional expedition • student design concept” label below.
4. Align the left edges and leave at least 90 px of breathing room. Use the title, supporting line and concept label to create three clear text levels.
5. Save the layered poster PSD and export a separate final JPEG. The campaign extension does not replace your core .blend, A/B renders or reflection evidence.

WHY The scene becomes a communication design when typography gives it a purpose and hierarchy. The label makes clear that the advertised expedition is fictional.

CHECK The title reads first, then the supporting line, while the balloon remains visible and the landscape is not covered by paragraphs.

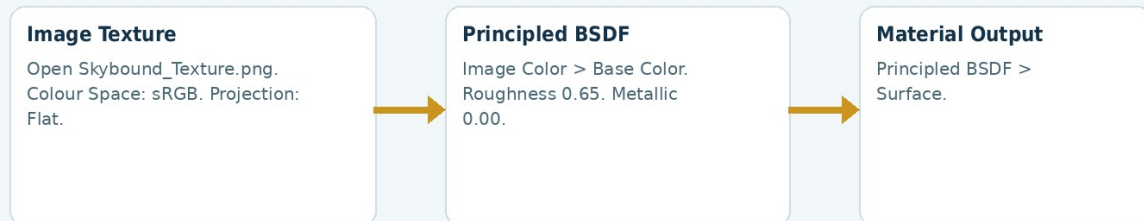
FIX Text competes with the main balloon? Shorten it and keep it in the empty left area. Use local contrast support rather than outlining every letter.

Tool references: P3 · Full links on pp. 54–55.

Surface: the connection and the wrap

07 / Connect colour — do not plug the image into Alpha

Teaching diagram • not a software screenshot



The existing UV map tells Blender where the image belongs.

Select Envelope first. A material on the basket will not decorate the balloon.

Material Preview shows the texture; Solid shading often does not.

08 / A flat image wraps around the envelope

Teaching diagram • not a software screenshot



Keep the texture dimensions and existing UV map.



Teaching diagrams, not software screenshots. Use the matching numbered steps for exact actions. Starting settings may need adjustment in the installed software.

Camera and light: keep them independent

11 / A simple, reproducible camera

Teaching diagram • not a software screenshot



Camera_Render

Location: X 0, Y -26, Z 6. Rotation: X 90°, Y 0°, Z 0°. Lens Type: Orthographic. Scale: 14.4.

Output

1200 × 1200 px. 100% for final; 50% for tests. Pixel aspect: 1:1. Full balloon and basket inside the frame.

12 / Key light first, gentle fill second

Teaching diagram • not a software screenshot

KEY — warm, upper left

Area / 1800 W / Size 5. Position (-6, -8, 13). Rotation (59.04°, 0°, -36.87°).



FILL — weaker, right

Area / 500 W / Size 6. Position (5, -4, 7). Rotation (90°, 0°, 51.34°).

Starting values for the supplied 12-unit model. Judge the rendered result.

Teaching diagrams, not software screenshots. Use the matching numbered steps for exact actions. Starting settings may need adjustment in the installed software.

Photoshop: local corrections and layer order

16 / Clip the correction to the balloon

Teaching diagram • not a software screenshot

BALLOON_LIGHT

Curves • clipped • black mask

BALLOON_TONE

Curves • clipped

BALLOON_RENDER

Embedded Smart Object

LANDSCAPE_SAT

Hue/Saturation • clipped

LANDSCAPE

Embedded Smart Object

18 / Final core layer order

Teaching diagram • not a software screenshot

GLOBAL_WARMTH

Photo Filter • affects the scene

FOG_FRONT

Painted pixel layer • not clipped

BALLOON_LIGHT

Curves • clipped to balloon

BALLOON_TONE

Curves • clipped to balloon

BALLOON_RENDER

Smart Object • your own PNG

LANDSCAPE_SAT

Hue/Saturation • clipped to landscape

LANDSCAPE

Smart Object • supplied photograph

Teaching diagrams, not software screenshots. Use the matching numbered steps for exact actions. Starting settings may need adjustment in the installed software.

When the result does not match

The balloon imports sideways or backward

Delete/undo that import. Use the prepared classroom OBJ with Forward Y and Up Z. Do not mix the raw original's axis assumptions with the prepared file.

There are many dozens of objects

You imported the raw original, or enabled the wrong split options. Use Balloon_Classroom.obj, Split by Object ON, Group OFF.

Only the fabric rotates

The controller does not parent every part. Undo the rotation, select all four meshes, make BALLOON_CTRL active last and parent with Keep Transform.

The camera or lights rotate with the balloon

They were parented by mistake. Clear Parent (Keep Transform) on those objects. They must remain outside BALLOON_CTRL.

The fabric is grey or pink

Grey: check Material Preview and Color → Base Color. Pink: the image is missing. Reopen the texture in the image node, then Pack Resources and save.

The texture wraps incorrectly

Use the retained Envelope UV map and full-size texture. Undo an accidental unwrap. Reimport the classroom model when the UVs have been changed beyond recovery.

The node graph is empty or says Background

Select Envelope and switch the Shader Editor to Object context, not World. Enable Use Nodes or create the envelope material once.

Rendered image is much darker than Material Preview

Material Preview may use studio lighting. Check your actual Area lights, camera and Rendered viewport. Adjust one light power at a time.

When the result does not match · 2

The render or preview is too slow

Save; use Solid shading while navigating. Test at 50% output. Avoid subdivision, volumetrics and extra lights. Teacher may test Cycles CPU when EEVEE is unsupported, but performance is device-dependent.

The saved PNG has an opaque background

Enable Film Transparent, rerender, then Image > Save As with PNG/RGBA. Check alpha over two colours before trying to erase the background.

The PNG contains only part of the balloon

Check Camera_Render framing and the render percentage. Do not crop the basket away. Keep identical full canvas dimensions for replacement renders.

A revised render changes size in Photoshop

Use the same 1200 × 1200 canvas and transparent margins. Replace Contents preserves transforms, but a differently cropped source changes apparent size.

The whole photo changes when I adjust the balloon

Place the correction directly above the balloon's clipping stack and use Layer > Create Clipping Mask. Global warmth is the intentional exception.

My fog hides the subject

Reduce fog, paint black on its mask and preserve the wordmark. Do not lower the entire balloon layer's opacity to make distance.

The scene is saved but my PNG is missing

Saving .blend does not save a standalone Render Result. Use Image > Save As from the render window and verify the exported file.

Progress or notes disappeared in the browser

Local storage is tied to a browser/profile and can be cleared or blocked. Import your JSON backup. Export notes after every class; the guide does not submit work automatically.

Explain your expedition

Name: _____ Class: _____ Date: _____

1 • 3D or 2D? What changed between your A and B renders? Explain why rotating a PNG would not produce that same change.

2 • Surface How did the UV map and Image Texture node work together? Identify the exact material input used.

3 • Light What was your strongest light? Describe one setting you changed and one visible result of that change.

4 • Alpha What does the A in RGBA mean? How did you verify your saved PNG actually contained transparency?

5 • Integration Which local adjustment or mask improved the composite most? Point to an area of the image and explain why.

6 • Revision What changed during the Blender-to-Photoshop round-trip, and did you keep or reject that revision?

7 • Support and credits List every prepared render, cut-out or mist support you used. Record the supplied source files and any additional image permissions.

Attach your own scene/node screenshots and A/B renders. A prepared support must be declared and cannot be presented as a self-rendered image.

Suggested evidence-based assessment

Five criteria, each scored 1–4; suggested total /20 only when evidence is sufficient. This is a proposed classroom rubric, not an official curriculum mapping. Apply school reporting policy for missing evidence.

Scene organisation and control

The four-part model imports correctly; BALLOON_CTRL transforms the complete model; the camera and lights remain independent. Evidence: .blend, hierarchy screenshot, brief live demonstration.

Material and lighting decisions

The learner connects the image texture to the correct material, preserves UVs and creates a coherent main light. Evidence: shader screenshot, saved scene, controlled lighting comparison.

Rendering and transparency

The learner makes two genuine 3D angles, frames the complete model and saves working PNG alpha. Evidence: self-rendered A/B PNGs and alpha check.

Photoshop integration

The learner uses Smart Objects, clipped corrections and restrained local atmosphere to connect the render to the photograph. Evidence: PSD, mask screenshot and final JPEG.

Explanation, revision and file care

The learner explains a visible revision, distinguishes 3D from flat-image changes, saves editable work and declares supports/sources. Evidence: reflection, .blend, PSD and organised exports.

A shared proficiency scale

4 Independent and intentional: accurate result, clear explanation and a justified refinement.

3 Secure: completes the core skill and explains it, with occasional minor prompting.

2 Developing: partial success or repeated prompting; explanation or evidence has gaps.

1 Beginning: attempts the skill but needs substantial guidance; result is not yet reliable.

0 / NE No usable evidence yet. Follow local reporting policy; absence of evidence is not automatically a permanent zero.

Do not grade extra effects, speed or support use as proxies for understanding. Collect a brief independent demonstration of navigation/parenting, a conversation about materials/light, and the saved product evidence.

Teacher preflight and pacing

Complete this before assigning the walkthrough independently. The HTML, PDF and source-model renders can be checked here; Blender/Photoshop operations still require a local software run.

- 1.** Install or confirm a school-approved Blender 4.5 LTS build and Photoshop desktop. Check local device requirements, GPU drivers, file permissions and student account access. Blender 5.2 users need a menu/settings spot-check; this guide deliberately targets 4.5.
- 2.** Import the classroom OBJ with Forward Y / Up Z and Validate Meshes. Confirm four objects, upright geometry, sensible normals and smooth shading. Keep the source archive unchanged.
- 3.** Connect Skybound_Texture.png to Envelope. Confirm the front lettering, UV seam and basket/rope appearance in the installed renderer.
- 4.** Build the parent controller, camera and lights from the settings card. Render a 600-pixel test; adjust light powers when needed. Make one 1200-pixel RGBA render and confirm save/reopen.
- 5.** Rotate the controller, rerender, and verify that the camera/lights remain fixed. Save a packed .blend, reopen it, and confirm the image texture survives.
- 6.** Place the rendered PNG in Photoshop and test Replace Contents with a same-size alternate render. Note any transparent-margin trimming and adapt the placement instruction visually.
- 7.** Test HTML opening, asset ZIP download, checkbox persistence and JSON backup/import in the actual student browser. Keep a PDF copy available. Notes are local, not a monitored classroom dashboard.
- 8.** Confirm image/model permission terms before public posting or redistribution. The uploads do not establish a new licence. Separate classroom use from publishing a public resource site.
- 9.** Decide what supports are allowed. A prepared PNG can support compositing, but it cannot stand in as evidence of importing, texturing or rendering. Provide an alternate live 3D demonstration opportunity.
- 10.** Before assessment, teach the required vocabulary and demonstrate selection, mouse navigation and saving. Use observations and short conversations, not only the finished poster.

Suggested pacing: Class 1 model/control; Class 2 texture/camera; Class 3 lights/renders; Class 4 Photoshop integration; Class 5 revision/explanation. Reserve a sixth class when required. Extensions follow the core.

Source integrity and permissions

Raw balloon geometry and materials

User-supplied balloon.obj and balloon.mtl. Header identifies a 3ds Max exporter and a 2014 creation date, not an author or licence. No map_Kd image reference is present in the original MTL.

Classroom model adaptation

Prepared from that model: coordinate conversion to Z-up, height 12 units, four named objects/materials, preserved envelope UVs, simplified supporting geometry and four short added suspension lines. This does not change the underlying source rights.

Original decal and reference

BalloonDecal.jpg is retained as Original_State_Fair_Decal.jpg. FinishedProject.png is retained as Original_Reference.png. They are references, not current real-event information.

Landscape and optional elements

OzLandscape.jpg is copied as Landscape.jpg. CityOutline.png, sunset-383072_1920.jpg and thunderstorm-567678_1280.jpg provide the optional city and clouds. Supplied files alone do not confirm public redistribution permission.

New lesson graphics

Skybound and blank texture layouts, instructional diagrams, painted mist and typography examples were created for this resource. No external font files are included.

Rendered supports and exemplars

Model images are renders of the supplied/adapted geometry in VTK 9.6.2. The scenes are programmatic source-photo composites. They are not AI-generated substitutes, Blender screenshots or proof that the exact Blender/Photoshop workflow has been executed.

Software verification

File parsing and source-model rendering were performed. HTML/PDF presentation checks are documented in QA_Report.md. Blender/Photoshop end-to-end operation remains a teacher preflight; no tested .blend project is supplied.

Original source files are retained in the pack where supplied. No new permission to publish someone else's photograph, model or artwork is implied. Confirm usage terms before a public GitHub or other public release.

Tool documentation

Primary documentation consulted for the workflow update. Lesson-specific numbers and design choices are authored starting points, not quoted settings from these sources. Accessed 17 September 2026.

A1 · Adobe — legacy Photoshop 3D removal notice

The July 2024 removal notice is relevant; its old Viewer suggestion is superseded by A2.

<https://helpx.adobe.com/photoshop/using/vanishing-point.html>

A2 · Adobe — removal of Substance 3D Viewer

Viewer and its Photoshop workflow were discontinued; do not install it for this lesson.

<https://helpx.adobe.com/substance-3d-viewer/desktop/getting-started/substance-3d-viewer-decommission.html>

B1 · Blender — 4.5 LTS release and support

Target version for this guide; not a claim that it is the newest Blender release.

<https://www.blender.org/releases/4-5/>

B2 · Blender Manual — Wavefront OBJ

OBJ/MTL relationship, axis conversion, splitting and mesh validation.

https://docs.blender.org/UATEST/manual/en/4.5/files/import_export/obj.html

B3 · Blender Manual — 3D View navigation

View navigation and framing; not object transformation.

<https://docs.blender.org/manual/en/4.5/editors/3dview/navigate/index.html>

B4 · Blender Manual — parenting objects

Parent relationships and keeping transforms.

https://docs.blender.org/manual/en/4.5/scene_layout/object/editing/parent.html

B5 · Blender Manual — shading objects

Shade Smooth changes surface shading, not topology.

https://docs.blender.org/manual/en/4.5/scene_layout/object/editing/shading.html

B6 · Blender Manual — Image Texture node

Image colour, alpha and active UV-map behaviour.

https://docs.blender.org/manual/en/4.5/render/shader_nodes/textures/image.html

B7 · Blender Manual — Principled BSDF

Roughness, metallic and material inputs.

https://docs.blender.org/manual/en/4.5/render/shader_nodes/shader/principled.html

Tool documentation • continued

Primary documentation consulted for the workflow update. Lesson-specific numbers and design choices are authored starting points, not quoted settings from these sources. Accessed 17 September 2026.

B8 • Blender Manual — cameras

Orthographic projection and framing.

<https://docs.blender.org/manual/en/4.5/render/cameras.html>

B9 • Blender Manual — colour management

View transforms; a consistent transform matters when comparing renders.

https://docs.blender.org/manual/en/4.5/render/color_management.html

B10 • Blender Manual — light objects

Area-light power, size and surface illumination.

https://docs.blender.org/manual/en/4.5/render/lights/light_object.html

B11 • Blender Manual — EEVEE Film

Transparent render background.

https://docs.blender.org/manual/en/4.5/render/eevee/render_settings/film.html

B12 • NASA — rendering an image preview in Blender

Render Result, PNG with RGBA and saving the image.

<https://science.nasa.gov/3d-resources/rendering-an-image-preview-in-blender/>

B13 • Blender Manual — packed data

Pack Resources is written when the .blend is saved.

https://docs.blender.org/UATEST/manual/en/4.5/files/blend/packed_data.html

P1 • Adobe — create Smart Objects

Place Embedded and replacing source contents.

<https://helpx.adobe.com/photoshop/using/create-smart-objects.html>

P2 • Adobe Learn — adjustment layers

Non-destructive local adjustments, clipping and masks.

<https://www.adobe.com/learn/photoshop/web/adjustment-layer>

P3 • Adobe — export files in Photoshop

Export As, image dimensions and output settings.

<https://helpx.adobe.com/photoshop/using/export-artboards-layers.html>